UPY

Embedded systems 6ºA

PCB manufacturing processes

Unit 3, 4

final project

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Abstract:

This report presents the second part of the project focused on the complete manufacturing process of a custom PCB designed for the STM32F411CEU6 microcontroller. The board is developed to support a wide range of current and future applications, including industrial-oriented features and expanded functionality beyond standard development boards. The design integrates the technical knowledge acquired throughout the author's academic program and aims to provide a robust platform for advanced embedded systems development. This document details the design rationale, manufacturing considerations, and the justification behind the selected components and layout decisions. The integration of this development board will enhance my skills for my major and will guide me through my job life.

Index terms

PCB, IPC standards, ICT, FT, THT, SMT, SMD.

Introduction

Previously, I developed a technical report describing the manufacturing process of my PCB design and how I placed an order for it through JLCPCB. Although that report covered the general structure and the overall layout of the board, it did not explain in detail how the circuit works or what the final functional scope of the design is.

In this second report, I will describe every relevant aspect of the PCB, beginning with the purpose of the project. Essentially, the board is intended to function as a development board like the WeAct Black Pill. However, it incorporates several additional features that enhance its capabilities. For example, this board can communicate directly with a computer without requiring an external USB-to-TTL converter, since it includes an integrated communication module, a feature not present in the original Black Pill design. This is only one of the several improvements implemented on the board; throughout this report, I will present all these features and explain how they contribute to the overall objective of the development platform.

Finally, the report will also justify the design decisions taken during the development of the PCB. This includes explanations about trace width selection, the choice between through-hole and SMT components, the application of various IPC guidelines, and other technical considerations. The goal is to provide a complete and transparent overview of the design process and the functional intent of the final PCB.

1. Project Definition and Justification

1.1. Problem Statement Modern agriculture faces the critical challenge of maximizing food production in limited spaces and controlled climates. Indoor farming systems depend entirely on artificial lighting to simulate the solar radiation necessary for photosynthesis. However, conventional lighting systems present significant limitations:

1. Static Spectrum: Most commercial fixtures offer a fixed light spectrum. However, plants require different wavelengths according to their biological stage (more blue light for vegetative growth and more red for flowering). The use of generic white light results in suboptimal photosynthetic efficiency.
2. Energy Inefficiency: Inadequate control of light intensity and timing generates considerable waste of electrical energy.
3. Lack of Remote Monitoring: The absence of connectivity in traditional controllers forces constant on-site supervision, hindering scalability and data collection for continuous crop improvement.

1.2. Proposed Solution To address these deficiencies, this project proposes the design and implementation of an IoT-enabled lighting system, specifically designed to optimize plant growth through precise control of the light spectrum, intensity, and timing.

The solution is based on the development of a custom PCB (Printed Circuit Board) that integrates the power stage and control logic onto a single board, eliminating the need for bulky external modules. The core of the system is the STM32F411CEU6 microcontroller, selected for its processing capacity to manage complex PWM signals and communication stacks.

Key technical features of the system include:

- Integrated Power Stage (Boost Converter): Design of a step-up (Boost) converter on the same PCB to power high-efficiency LED arrays, ensuring constant and stable current.
- IoT Connectivity (2.4 GHz): Implementation of a wireless communication module to allow variable monitoring and remote lighting control.
- Dynamic Spectrum Modulation: Independent control of red and blue light channels, allowing the spectral ratio to be adjusted in real-time to match the phenological stages of the plant (germination, vegetation, or flowering).

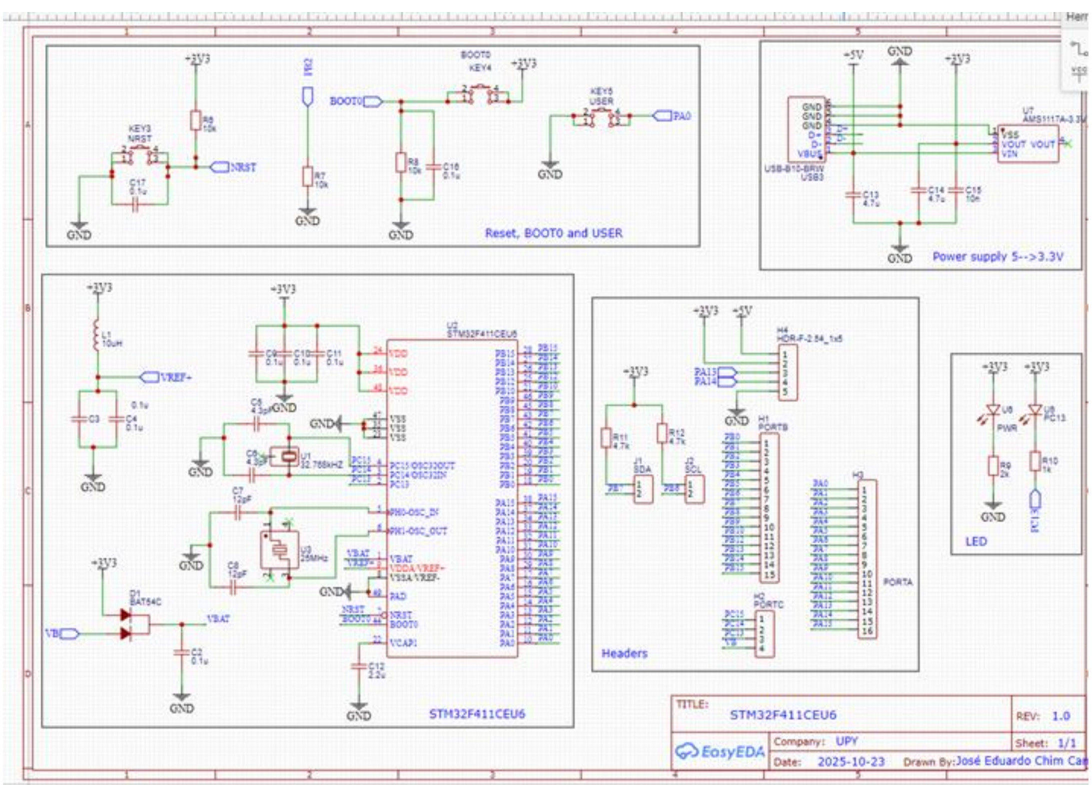
This technology aims directly to improve the biomass obtained and reduce cultivation times, promoting more sustainable and energy-efficient agriculture.

2. Electronic and Normative Design

2.1. System Overview The proposed system is an embedded development platform based on the STM32F411CEU6 microcontroller. Unlike standard commercial development boards (such as the Black Pill), this design natively integrates IoT connectivity and a power stage, eliminating the need for external modules and "shields."

The schematic design is divided into four main functional blocks:

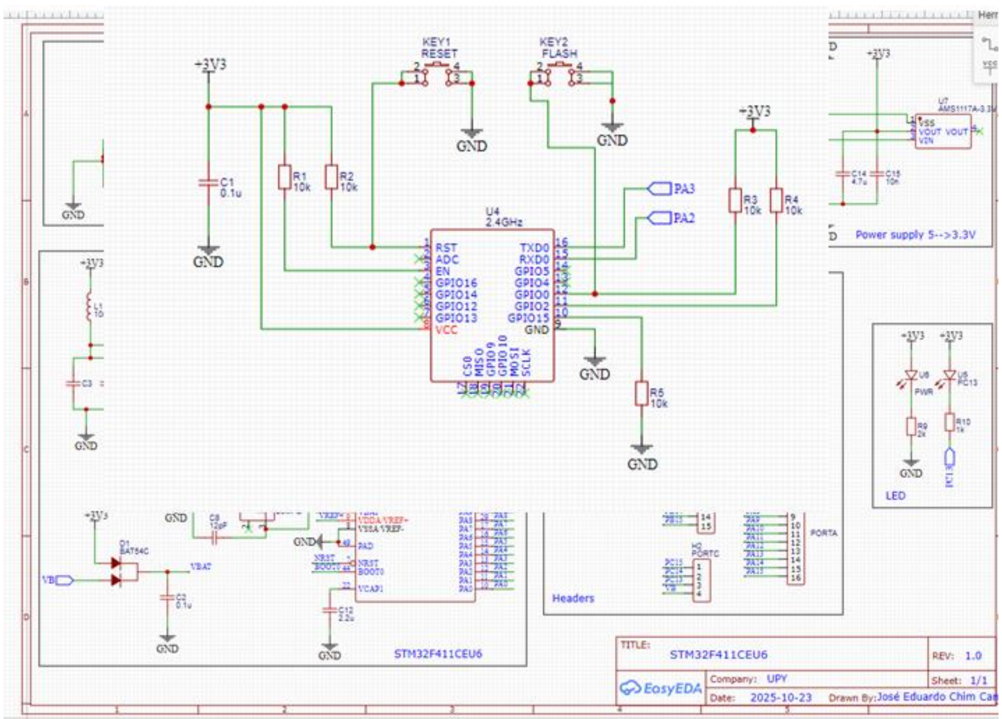
2.2. Control Subsystem (MCU)



The minimum system necessary for the operation of the STM32F411CEU6 was implemented (See Figure 2.1).

- Clock and Reset: A 25 MHz crystal oscillator is included for the main clock and a 32.768 kHz oscillator for the RTC (Real Time Clock). The reset circuit (NRST) ensures a stable startup.
- Decoupling: Decoupling capacitors (100nF and 4.7\mu F) were placed on the power supply pins (VDD) to filter high-frequency noise, in accordance with the manufacturer's recommendations (STMicroelectronics).

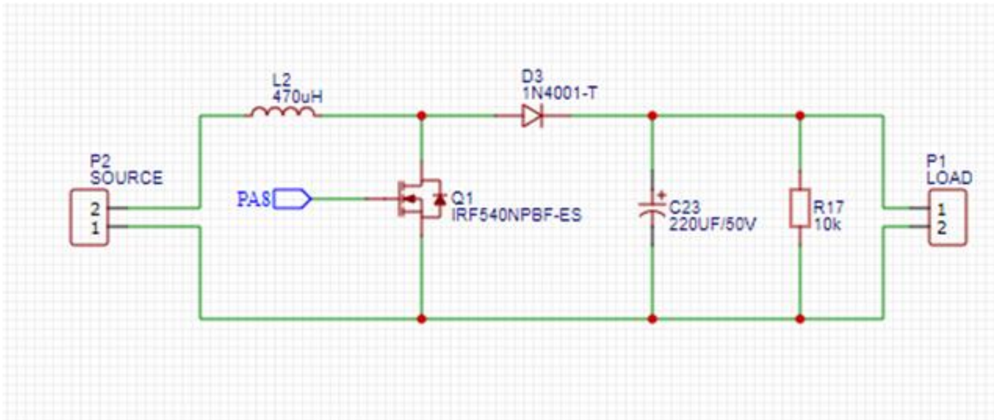
2.3. IoT Connectivity Module and Communication Selection



To address the lack of connectivity in standard boards, the ESP8266 Wi-Fi module (model ESP-12F) was integrated.

- UART Interface: Communication between the STM32 and the ESP8266 is established via a serial port.
- Bus Selector (Switch): A switching circuit was designed using jumpers/switches, allowing the user to select the communication path (direct USB to PC or internal MCU-ESP8266 communication) or disable it entirely to save power. This provides superior versatility for debugging and firmware uploading.

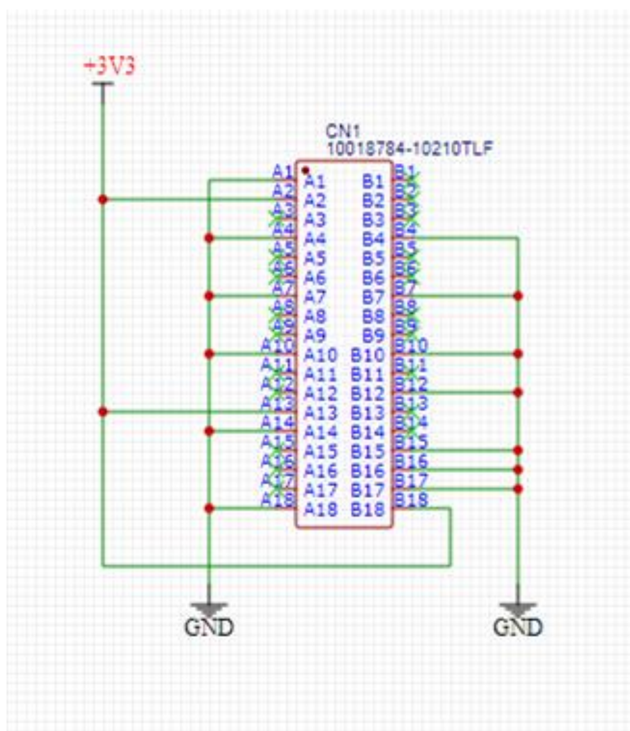
2.4. Power Stage: Boost Converter



To fulfill the objective of controlling higher voltage loads (such as 12V or 24V DC fans), a Boost (Step-up) DC-DC converter topology was integrated into the board.

- Operation: The STM32 generates a PWM signal via pin PA8 (Timer 1, Channel 1). This signal switches the IRF540 MOSFET, allowing the input voltage (5V USB) to be boosted up to a maximum of 20V.
- Application: This stage enables temperature-based fan speed control, managed by both local logic and remote commands via IoT.

2.5. Expansion Interface (PCIe Form Factor)



A PCIe x1 edge connector (model 10018784-10210TLF) was included.

- Technical Justification: Since the STM32F411 does not feature a controller for the PCI Express protocol, this connector is used exclusively as a robust mechanical interface and signal expansion (Breakout). It allows the board to be connected to a "backplane" or master base in a secure and standardized manner, leveraging the availability and reliability of commercial PCIe connectors to distribute GPIO and power signals.

2. PCB Design and Manufacturing

2.1. Component Selection and Assembly Strategy For the physical design of the printed circuit board (PCB), a mixed-technology architecture was chosen, combining Surface Mount (SMD) and Through-Hole (THT) components. This decision is driven by criteria of mechanical robustness, thermal management, and ease of manufacturing and rework.

A. THT Components (Through-Hole Technology) Through-hole packages were selected for the power stage of the Boost converter, connection headers, and the USB Type-B port.

- Mechanical Justification: Connectors (USB and headers) are subjected to constant mechanical stress due to cable connection and disconnection. THT anchoring provides superior resistance to detachment compared to their SMD counterparts.
- Thermal Justification: In the Boost converter stage, components such as inductors and diodes handle high currents. THT packages offer greater thermal mass and better passive heat dissipation, which is crucial for system stability without the need for complex external heatsinks.
- Availability: The availability of spare parts from local suppliers was prioritized to facilitate maintenance and rapid prototyping iteration.

B. SMD Components (Surface Mount Technology) For the rest of the logic and control system, surface mount components were used to optimize PCB area and improve signal integrity.

- Resistors (1206 Package): The 1206 size was chosen instead of smaller formats (0402/0603) for medium power resistors. This format offers an ideal balance between power dissipation and ease of handling for manual soldering during the prototyping phase.
- Capacitors (0603 Package): The use of 0603 was standardized for the decoupling network. Its reduced size allows for placement very close to the MCU power pins, minimizing parasitic inductance and noise on the voltage lines.
- Microcontroller (UFQFPN-48): The STM32F411CEU6 was implemented in its QFN (Quad Flat No-leads) package. This format is critical for the design due to its low pin inductance and, primarily, for its exposed thermal pad on the bottom, which is soldered directly to the PCB ground plane to maximize processor heat dissipation.
- Voltage Regulator (SOT-223): For the linear regulator (LDO), the SOT-223 package was preferred over the SOT-23. Its larger copper contact area allows for efficient dissipation of the heat generated by the voltage drop (from 5V to 3.3V) under load.

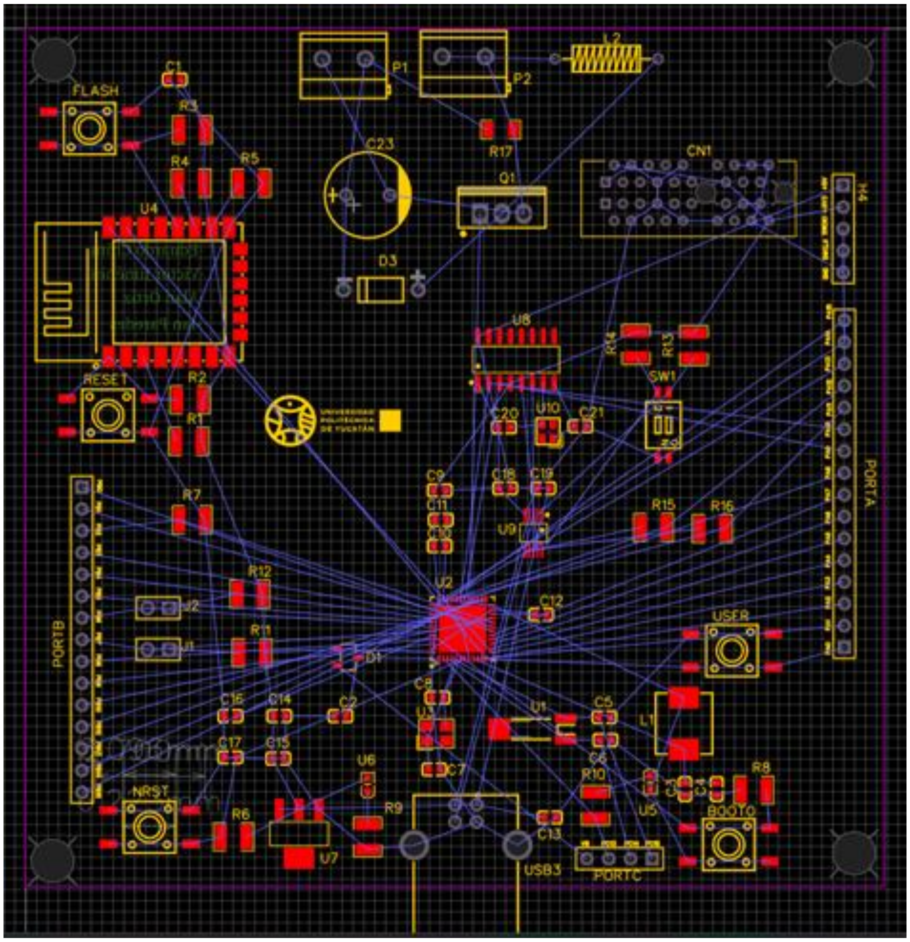
2.2. Bill of Materials (BOM)

Name	Designator	Footprint	Quantity
470uH	L2	IND-TH_BD3.0-L8.0-P12.00-D0.6	1
PORTB	PORTB	HDR-F-2.54_1X15	1
PORTC	PORTC	HDR-F-2.54_1X4	1
PORTA	PORTA	HDR-F-2.54_1X16	1
HDR-F-2.54_1x5	H4	HDR-F-2.54_1X5	1
RESET	RESET	KEY-SMD_4P-L6.0-W6.0-P3.90-LS10.0	1
FLASH	FLASH	KEY-SMD_4P-L6.0-W6.0-P3.90-LS10.0	1
NRST	NRST	KEY-SMD_4P-L6.0-W6.0-P3.90-LS10.0	1
BOOT0	BOOT0	KEY-SMD_4P-L6.0-W6.0-P3.90-LS10.0	1
USER	USER	KEY-SMD_4P-L6.0-W6.0-P3.90-LS10.0	1
220UF/50V	C23	CAP-TH_BD10.0-P5.00-D1.0-FD	1
1N4001-T	D3	DO-41_BD2.4-L4.7-P8.70-D0.9-RD	1
LOAD	P1	CONN-TH_2P-P5.00	1
SOURCE	P2	CONN-TH_2P-P5.00	1
IRF540NPBF-ES	Q1	TO-220-3_L10.0-W4.6-P2.54-L	1
10k	R17	R1206	1
330	R13,R14	R1210	2
10	R15,R16	R1210	2
CHS-02TB1	SW1	SW-SMD_2P-L5.4-W4.1-P1.27-LS8.0	1
0.1u	C1,C2,C3,C4,C9,C10,C11,C16,C17	C0603	9
4.3pF	C5,C6	C0603	2
12pF	C7,C8	C0603	2
2.2u	C12	C0603	1
4.7u	C13,C14	C0603	2
10n	C15	C0603	1
100nF	C18,C19	C0603	2
22pF	C20,C21	C0603	2
BAT54C	D1	SOT-23-3_L2.9-W1.6-P1.90-LS2.8-BR	1
SDA	J1	HDR-M-2.54_1X2	1
SCL	J2	HDR-M-2.54_1X2	1
10uH	L1	IND-SMD_L7.1-W6.6_MJC-0603T	1
10k	R1,R2,R3,R4,R5,R6,R7,R8	R1210	8

2k	R9	R1210	1
1k	R10	R1210	1
4.7k	R11,R12	R1210	2
32.768kHZ	U1	CRYSTAL-SMD_32768HZ-CRYSTAL_2-6SMD	1
STM32F411CEU6	U2	UFQFPN-48_L7.0-W7.0-P0.50-BL-EP	1
25MHz	U3	CRYSTAL-SMD_4P-L3.2-W2.5-BL	1
2.4GHz	U4	WIFIM-SMD_ESP-12F-ESP8266MOD	1
PC13	U5	LED0603-RD_1	1
PWR	U6	LED0603-RD_1	1
AMS1117A-3.3V	U7	SOT-223_L6.3-W3.5-P2.30-LS7.0-BR	1
CH340G	U8	SOIC-16_L9.9-W3.9-P1.27-LS6.0-BL	1
FSUSB42MUX	U9	MSOP-10_L3.0-W3.0-P0.50-LS5.0-BL	1
12MHz	U10	CRYSTAL-SMD_4P-L2.5-W2.0-BL_XTAL12MHZ	1
USB-B10-BRW	USB3	USB-B-TH_USB-B10-BRW	1
10018784-10210TLF	CN1	CONN-TH_10018784-10210TLF	1

2.3. Layout Design and Placement Strategy

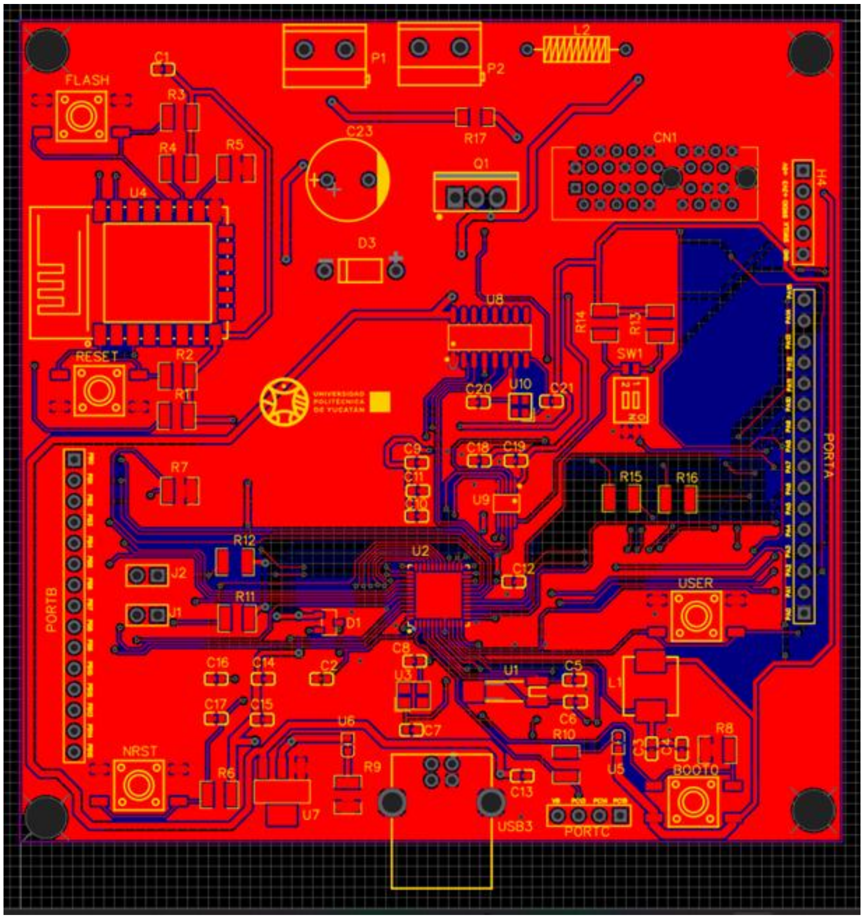
Upon completion of the schematic, the physical design (Layout) phase was initiated within the EDA environment. The initial transition generates "ratlines" (logical connection lines without physical routing), presenting the challenge of organizing the components within the defined area of the board



Placement Strategy: Component placement was not arbitrary; it adhered to functional criteria and signal integrity requirements:

- IoT Module (ESP8266): Positioned on the left edge of the PCB, ensuring the integrated antenna remained free of copper and ground planes underneath. This layout is critical to prevent RF signal shielding and maximize Wi-Fi range.
- User Connectors: USB ports, screw terminals (for load and source), and expansion headers were placed along the perimeter edges to facilitate mechanical connection and end-user access.
- Power Integrity: Decoupling capacitors (0603) were placed as close as possible to the STM32F411 power pins (VDD), minimizing the inductance loop to effectively filter high-frequency noise.

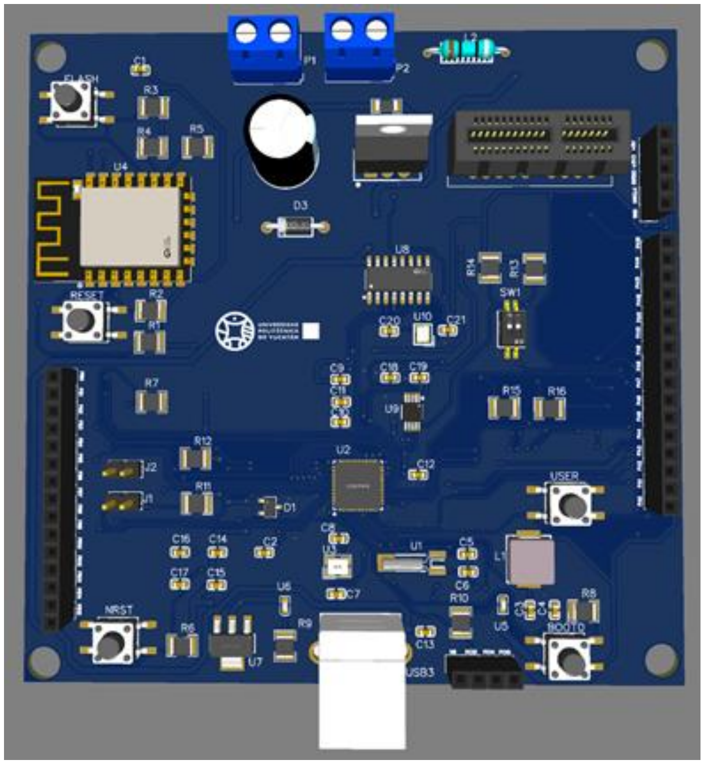
2.4. Routing and IPC-2221 Standard Trace routing was performed on a two-layer board (Top and Bottom). To determine trace width, the criteria of the IPC-2221 standard (Generic Standard on Printed Board Design) were applied.



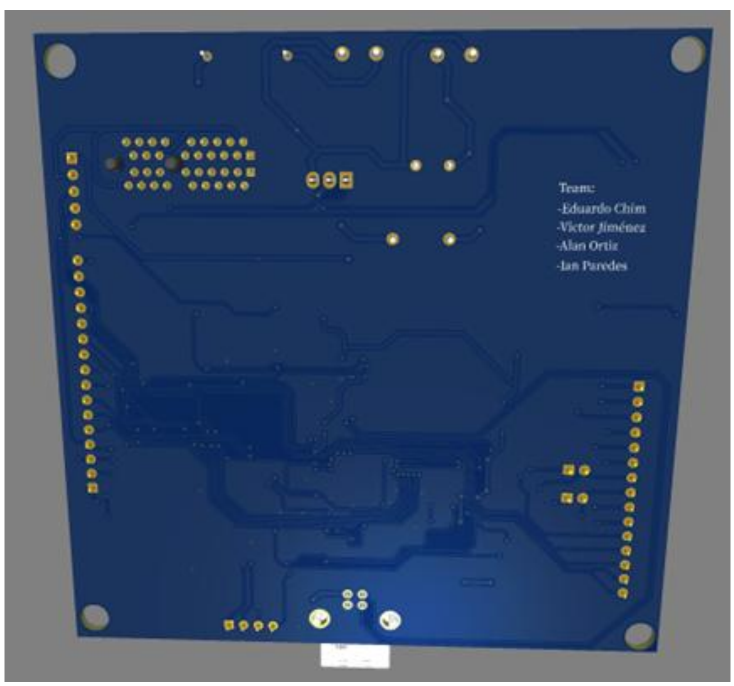
- Signal Traces: For data lines (GPIO, SPI, UART), standard widths (approx. 10-12 mils) were used, which are sufficient for low-current digital signal transmission.
- Power Traces: In the Boost converter section and main power lines, trace width was significantly increased to handle the required current without dangerous temperature rises. (Note: Detailed ampacity and temperature calculations will be presented in the supplementary analysis section) .
- Trace Geometry: The use of 90-degree angles in traces was strictly avoided, utilizing 45-degree bends instead. This reduces the possibility of "acid traps" during manufacturing and slightly improves impedance in high-speed signals.

2.5. Reference Planes and Manufacturing

As observed in the 2D view of the final design, Ground Planes were implemented on both sides of the PCB (represented in red for Top and blue for Bottom).



On the bottom layer (Bottom Silk Layer), the team members' names were included via silkscreen, ensuring project identification during the manufacturing phase



- Thermal and Electrical Benefits: These planes act as heat sinks for power components and provide a low-noise reference for the microcontroller, reducing electromagnetic interference (EMI).

- Stitching Vias: Multiple vias were added connecting the top plane to the bottom one to ensure equipotentiality throughout the board and avoid long return loops[2].
- Manufacturing Files: Finally, the design was verified through a Design Rule Check (DRC) to ensure there were no shorts or minimum clearance violations, successfully generating the Gerber files for professional production[3].

2.6. IPC Standards Implementation in Design

To ensure assembly reliability (DFM) and signal integrity, the design strictly adhered to the following industry standards4:

2.6.1. Footprint Design (IPC-7351C)

Land Patterns were defined by selecting the appropriate "Tolerance Class" according to component criticality5:

- 0603 Passive Components (Nominal Level): The nominal tolerance (Class B) was applied with pads of 0.80mm x 0.75mm. This ensures a robust solder fillet and minimizes the risk of "tombstoning" during reflow, complying with Chapter 5 of the standard[6].
- STM32F411 Microcontroller (Minimum Level): For the fine-pitch (0.5mm) QFN-48 package, the "Minimum" tolerance (Class A) was used. This reduces the pad width to 0.25mm to maximize isolation distance and avoid solder bridging. A 4x4 via array was included in the central thermal pad according to Section 8.3 of IPC-7351C[7].
- ESP-12F Module (Maximum Level): Due to the mass and size of the module, the "Maximum" tolerance (Class C) was used, enlarging the pads to improve mechanical resistance against vibrations[8].

Component	Package	Pad (L x W)	Tolerance	Technical Justification
Capacitor	0603	0.8 x 0.75 mm	Nominal	Optimizes self-alignment during reflow.
Resistor	1206	1.6 x 1.55 mm	Nominal	Robust area for medium thermal dissipation.
STM32F411	QFN-48	0.5 x 0.25 mm	Minimum	Prevents bridging on fine-pitch components.

Component	Package	Pad (L x W)	Tolerance	Technical Justification
AMS1117	SOT-223	1.6 x 1.2 mm	Maximum	Maximizes copper area for thermal dissipation.
ESP-12F	Wi-Fi Module	1.5 x 1.0 mm	Máximum	Additional mechanical support for module weight.

2.6.2. Trace Width Calculation (IPC-2221) Traces were dimensioned by calculating the allowable temperature rise

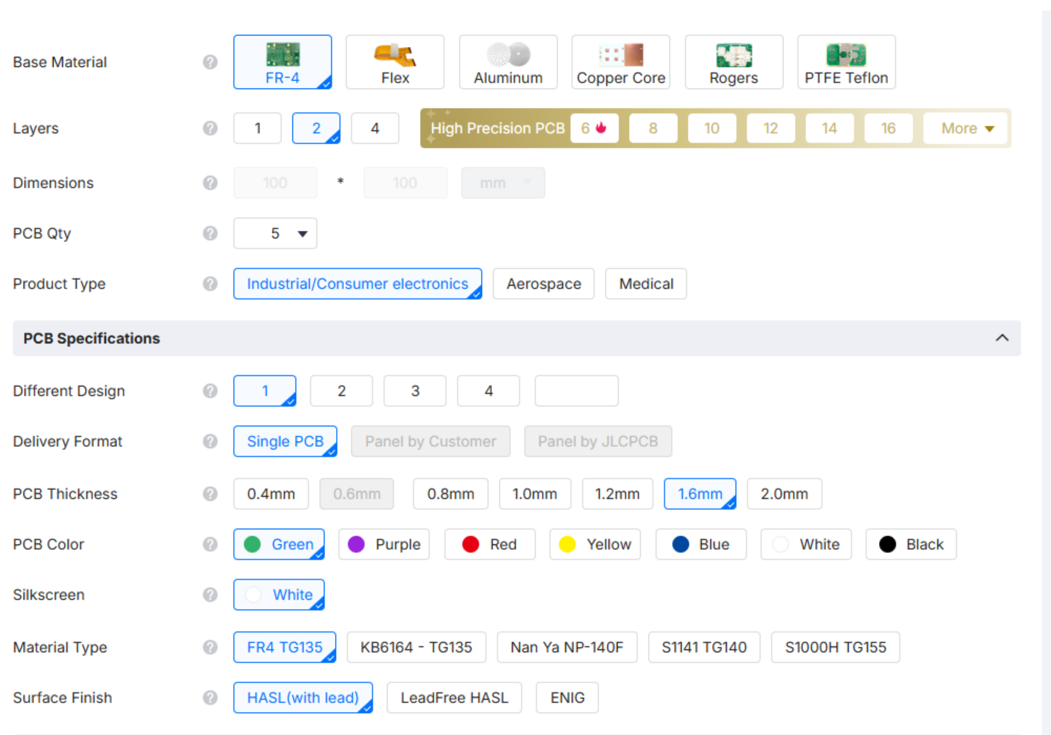
Table 2.2: Summary of Land Patterns (IPC-7351)

Use/signal	Estimated current (A)	Calculated width (mm)	Chosen width (mm)	Justification
Voltage supply 3.3 and 5V	0.5 - 1 A	0.305 - 0.78 mm	0.8 mm	This width is more than enough for all the components in the PCB.
GPIO (STM32F411CEU6, ESP8266, FSUSB42MUX and CH340G)	0.01 - 0.025 A	0.001- 0.004 mm	0.15 mm	The width for these traces are very small, and sometimes the manufacturer can´t do the exactly width that we want, thus I chose this width because it is small but more than enough for the GPI
D+, D-	0.05 A	0.0125 mm	0.2 mm	For a standard 1.6mm FR4 board, the width for a 22Ω differential pair is often around 0.2mm
MOSFET	1 A	0.78 mm	0.8 mm	Provides a large safety margin, very low heat, and is highly robust.

Connectors	1 A	0.78 mm	0.8 mm	Just like the MOSFET, it is more than enough for the boost converter.
Resistors, capacitor and inductors	0.01- 0.025 A	0.001 - 0.004 mm	0.15 mm	This is the width that we are going to implement for small traces. This width is useful for all the applications that we can make in the PCB.
Potential part	1 A	0.78 mm	0.8 mm	This applies for the diode and the capacitor used in the part of the boost converter

2.7. Quotation and Manufacturing Viability

The PCB order was placed with JLCPCB using the specifications detailed below. To maintain a cost-effective approach, standard base materials and manufacturing parameters were selected where possible.



High-spec Options

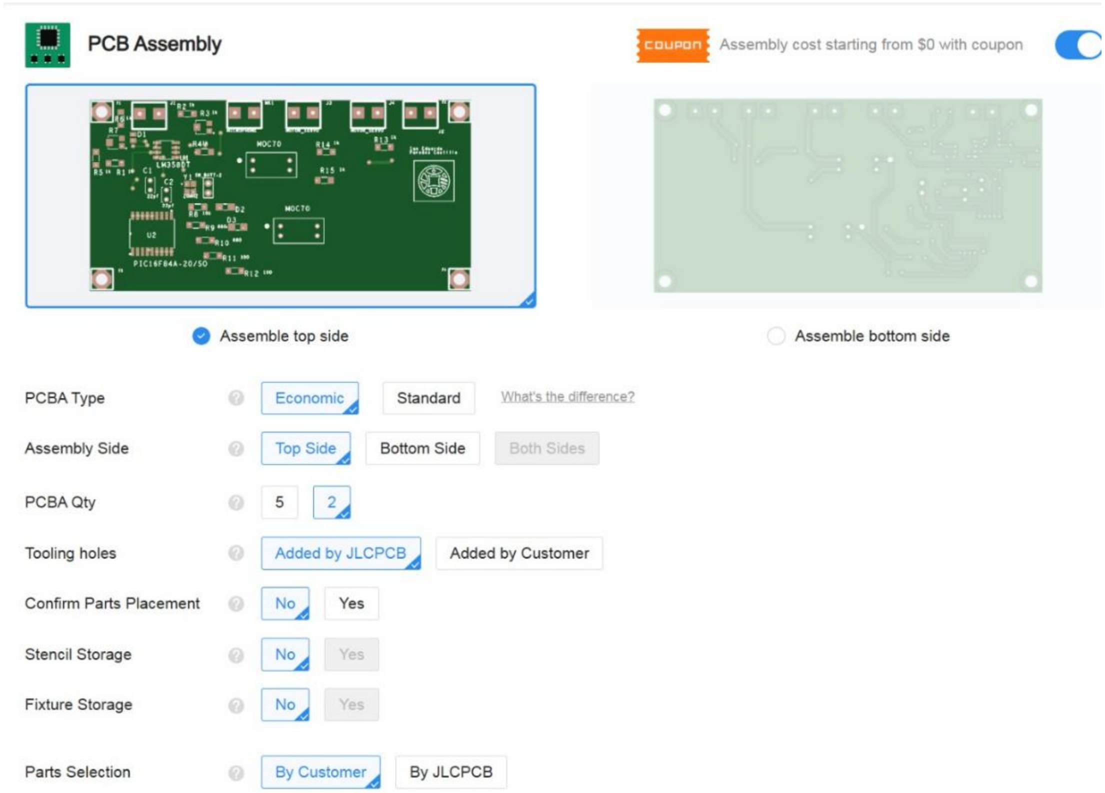
Outer Copper Weight	1 oz / 2 oz
Via Covering	Tented / Untented / Plugged / Epoxy Filled & Capped / Copper paste Filled & Capped
Min via hole size/diameter	0.3mm/(0.4/0.45mm) / 0.25mm/(0.35/0.4mm) / 0.2mm/(0.3/0.35mm) / 0.15mm/(0.25/0.3mm)
Board Outline Tolerance	±0.2mm(Regular) / ±0.1mm(Precision)
Confirm Production file	No / Yes
Mark on PCB	Remove Mark / Order Number(Specify Position) / 2D barcode (Serial Number) / Order Number
Electrical Test	Flying Probe Fully Test
Gold Fingers	No / Yes
Castellated Holes	No / Yes
Edge Plating	No / Yes
Blind Slots	No
UL Marking	No / Yes (Any Position) / Yes (Specify Position)

Only specified brands' materials (e.g., Nan Ya NP-140F) support adding UL markings.

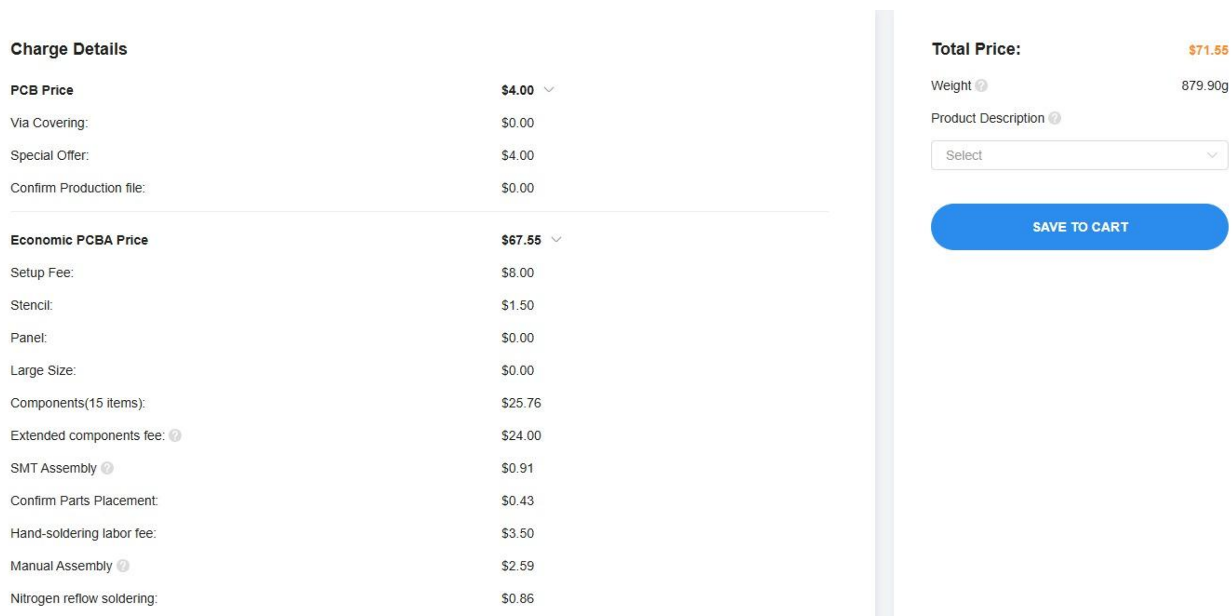
Advanced Options

4-Wire Kelvin Test	No / Yes
Paper between PCBs	No / Yes
Appearance Quality	IPC Class 2 Standard / Superb Quality
Silkscreen Technology	Ink-jet/Screen Printing Silkscreen / High-precision Printing Silkscreen / EasyEDA multi-color silkscreen / High-definition Exposure Silkscreen
Package Box	With JLCPCB logo / Blank box
Inspection Report	No / Final Inspection Report / Electrical Test Report / ROHS Test Report

PCB Remark



This is the total price for the PCB



Charge Details

PCB Price	**$4.00**
Via Covering:	$0.00
Special Offer:	$4.00
Confirm Production file:	$0.00
Economic PCBA Price	**$67.55**
Setup Fee:	$8.00
Stencil:	$1.50
Panel:	$0.00
Large Size:	$0.00
Components(15 items):	$25.76
Extended components fee:	$24.00
SMT Assembly	$0.91
Confirm Parts Placement:	$0.43
Hand-soldering labor fee:	$3.50
Manual Assembly	$2.59
Nitrogen reflow soldering:	$0.86

Total Price:	$71.55
Weight	879.90g
Product Description	Select

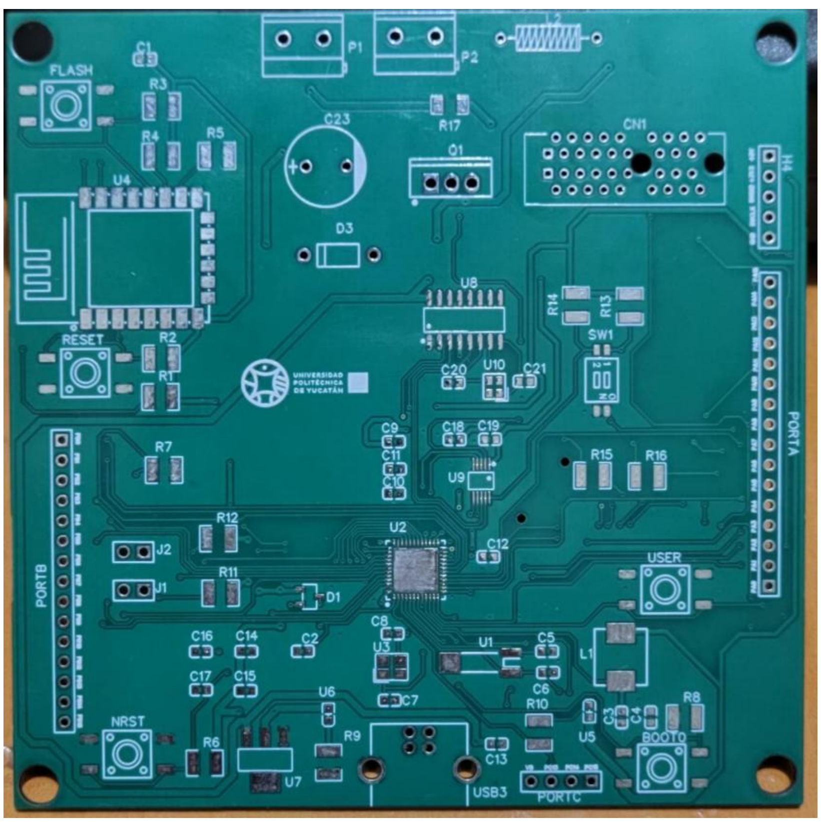
SAVE TO CART

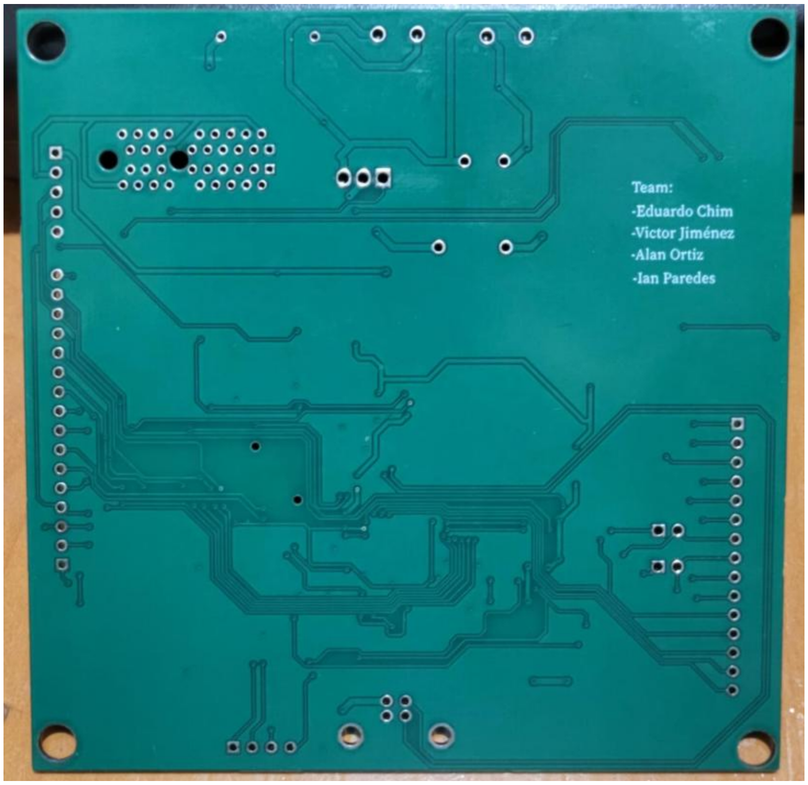
The price is high but because I put the PCB will come with some components already soldered. This naturally increases the manufacturing cost, as JLCPCB must perform additional assembly processes, including pick-and-place, soldering, testing, and verification. Even so, analyzing the quotation allows us to evaluate both the economic and technical viability of the design.

From an economic perspective, the cost remains reasonable considering the complexity of the board, the number of surface-mount components, and the added value of automated assembly. Producing the PCB on our own or through less specialized services would increase the risk of errors and rework, resulting in higher long-term expenses. JLCPCB's pricing structure is competitive and transparent, which makes it suitable for prototypes, student projects, and even small-scale production.

From a technical standpoint, the PCB is fully viable. JLCPCB supports standard FR-4 materials, the thickness required for our traces, precise drilling tolerances, and the component footprints used in the design. Their DFM tools help ensure that the PCB meets industry guidelines, reducing the probability of manufacturing failures. Additionally, using a professional assembly service guarantees that critical components such as the STM32 microcontroller, regulators, and fine-pitch ICs are placed accurately and reliably.

3.1. Incoming PCB Inspection (IPC-A-600)





Before proceeding with assembly, the quality of the Bare Board was verified under the IPC-A-600 (Acceptability of Printed Boards), Class 2 standard.

- Annular Ring Integrity: It was verified that the vias (0.3mm drill) maintained a copper annular ring greater than 0.050mm, ensuring connectivity between layers despite drilling tolerances.
- Solder Mask Registration: A mask expansion (pullback) of 0.05mm was confirmed, ensuring that no mask encroached upon the SMD pads (Target Condition).
- Laminate Condition: Visual inspection confirmed the board was free of measling (white spots) or delamination, verifying that the FR-4 material did not suffer excessive thermal stress during fabrication.

(Insert photos of your bare PCB here, preferably close-ups)

3.2. Assembly Quality Control (IPC-A-610)

Once soldering was completed, the final PCBA was audited under the IPC-A-610 Class 2 (Dedicated Service Electronic Products) standard. This classification is appropriate for systems requiring high reliability and continuous service.

4. Functional System Implementation

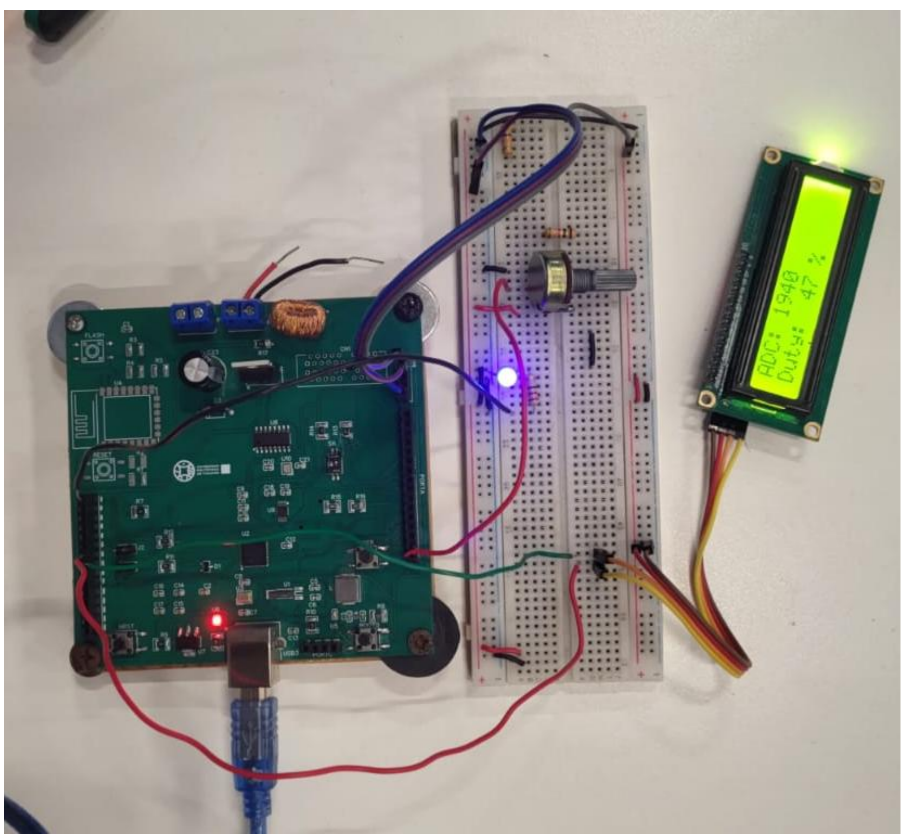
4.1. Integration and Experimental Setup To validate the joint operation of hardware and software blocks, an experimental test bench was configured (See Figure 4.1). The objective was to verify the PCB's capability to acquire real-world signals, process them, and act accordingly.

The integrated system configuration consists of:

- Input (Sensors): A potentiometer connected to the STM32 ADC port (Pin PA0) simulating a temperature sensor or user input.
- Processing: The STM32F411 performs the analog reading and calculates a proportional Duty Cycle.
- Local Output (HMI): A 16x2 LCD screen connected via I2C displaying real-time values, and a blue LED visually indicating the PWM signal intensity.
- Telemetry: Simultaneous data transmission to the PC via UART.

4.2. I/O Validation (ADC and LCD) The functionality of the expansion headers and the I2C communication bus was verified. As evidenced in Figure 4.1, the system is operational:

- ADC Reading: The display shows a raw value of ADC: 1940 (12-bit resolution).
- Control Logic: The microcontroller correctly calculates the power percentage, displaying Duty: 47%.
- Visual Indicator: The blue LED on the breadboard lights up with an intensity corresponding to 47%, confirming that the PWM signal is correctly outputting from the PCB to the outside world.

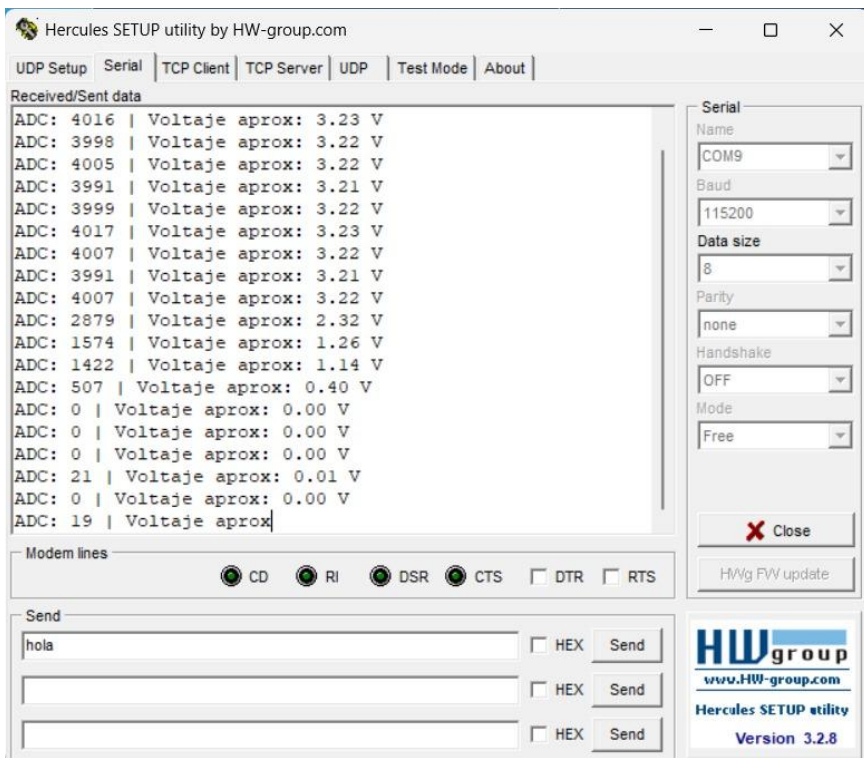


4.3. Communication and Telemetry Validation (UART)

To confirm the remote monitoring capability (a requirement for future IoT implementation), the system was connected to the computer's USB port using the board's integrated CH340G bridge. The Hercules SETUP utility software was used to capture the data frame.

Test Results

- Data Transmission: The system sends a formatted frame that includes the raw ADC value and its conversion to floating-point voltage.
- Precision: Dynamic variation of values is observed (from 3.23 V down to 0.00 V), demonstrating that the STM32 is continuously sampling and transmitting without perceptible latency at 115200 baud.
- Integrity: No "garbage" characters or parity errors are observed, validating the design of the differential D+/D- traces and the stability of the oscillator.



1. 4.4. Integration Results Matrix The following table summarizes the functional tests performed on the final assembled prototype, confirming compliance with design requirements.
2. Table 4.1: Functional Validation Matrix

Tested Module	Test Performed	Observed Result	Status
Power Supply	Red LED (PWR) ON and 3.3V stability	System stable, no reboots.	SUCCESS
ADC (Sensor)	Potentiometer Variation (0V - 3.3V)	Reading from 0 to 4095 detected in software.	SUCCESS
Actuators (PWM)	External LED intensity control	Variable brightness proportional to ADC.	SUCCESS
I2C Interface	Writing to 16x2 LCD Display	Sharp characters, fluid update.	SUCCESS
Serial Communication	Data sending to PC (Hercules)	Readable and correctly formatted data.	SUCCESS

5. Firmware Development and Logic Control

5.1. Development Environment and Architecture The firmware was developed using the STM32CubeIDE ecosystem, employing HAL (Hardware Abstraction Layer) libraries to ensure portability and facilitate code maintenance. The language used was standard C.

The software architecture is based on an infinite main loop (Super Loop) that manages data acquisition and peripheral updates sequentially, ensuring adequate deterministic latency for the system's thermal time constant.

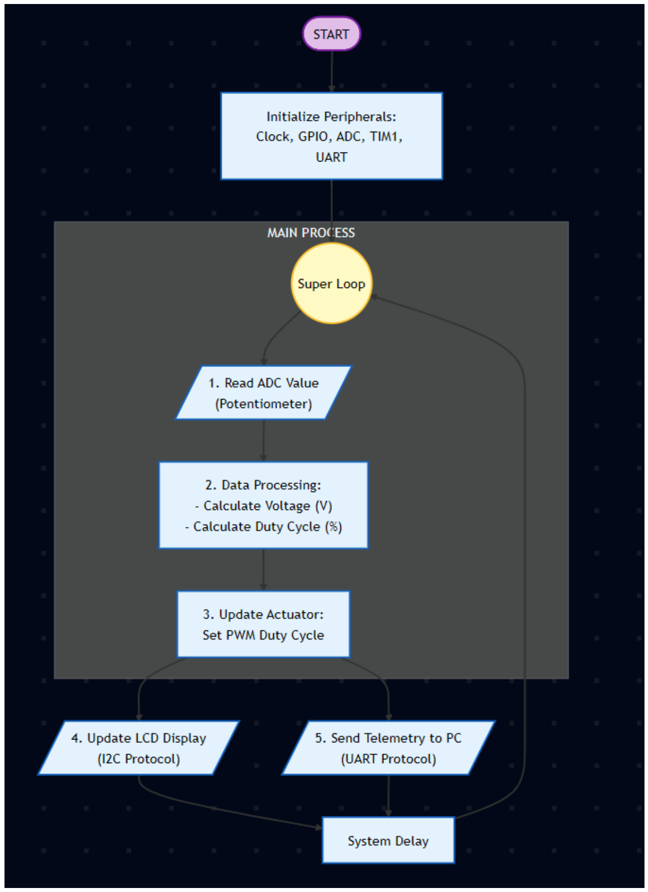
5.2. Peripheral Configuration (Hardware Domain) To achieve the demonstrated functionality, the following internal modules of the STM32F411CEU6 were configured:

- ADC1 (Analog-to-Digital Converter):
  - Resolution: 12 bits (Values from 0 to 4095).
  - Mode: Continuous polling.
  - Channel: PA0, configured to read the voltage from the voltage divider potentiometer.
- TIM1 (General Purpose Timer):
  - Function: PWM signal generation.
  - Configuration: Prescaler and Period adjusted to obtain a safe switching frequency for the MOSFET and the Boost circuit.
  - Output: Channel 1 (PA8) mapped to the Transistor Gate.
- USART1 (Serial Communication):
  - Baud Rate: 115200 bps.
  - Format: 8 data bits, no parity, 1 stop bit (8N1).

  - Usage: Telemetry transmission (ADC Values and Calculated Voltage) to the PC.
- I2C1 (Inter-Integrated Circuit):
  - Mode: Standard Master (100 kHz).
  - Usage: Control of the 16x2 LCD display via the PCF8574 I/O expander.

```c
while (1)
{
    HAL_ADC_Start(&hadc1);
    HAL_ADC_PollForConversion(&hadc1, HAL_MAX_DELAY);
    valor_adc = HAL_ADC_GetValue(&hadc1);
    HAL_ADC_Stop(&hadc1);

    // 1. Convertir ADC a porcentaje (0-100%)
    porcentaje_duty = ((float)valor_adc / 4095.0f) * 100.0f;

    // 2. Convertir porcentaje a CCR3 (0-ARR)
    uint32_t arr = __HAL_TIM_GET_AUTORELOAD(&htim3);
    uint32_t duty_counts = (uint32_t)((porcentaje_duty / 100.0f) * arr);

    htim3.Instance->CCR3 = duty_counts;

    // 3. Mostrar en LCD
    sprintf(buffer_adc, "ADC: %4u", valor_adc);
    sprintf(buffer_volt, "Duty: %3.0f %%", porcentaje_duty);

    HD44780_Clear();
    HD44780_SetCursor(0,0);
    HD44780_PrintStr(buffer_adc);
    HD44780_SetCursor(0,1);
    HD44780_PrintStr(buffer_volt);
    sprintf(buffer_serial, "ADC=%u  Duty=%.1f%%\r\n", valor_adc, porcentaje_duty);
    HAL_UART_Transmit(&huart1, (uint8_t*)buffer_serial, strlen(buffer_serial), HAL_MAX_DELAY);

    HAL_Delay(200);
  }
```

5.3. Diagrama de Flujo del Programa



6. Testing and System Validation

6.1. Protocol test

To validate the integral operation of hardware and software, stress and linearity tests were conducted under laboratory conditions (ambient temperature 25^{\circ}C).

6.2. ADC Linearity Validation

The precision of the STM32 Analog-to-Digital Converter was analyzed by comparing the obtained digital value (UART telemetry) against the expected voltage.

Using the data captured in the serial monitor (See Section 4), the following linearity table was constructed:

Table 6.1: ADC Linearity Test (12-bit Resolution)

Physical Input	ADC Value (Raw)	Calculated Voltage (MCU)	Real Voltage (Multimeter)	Absolute Error
Minimum	0	0.00 V	0.00 V	0.00 V

Physical Input	ADC Value (Raw)	Calculated Voltage (MCU)	Real Voltage (Multimeter)	Absolute Error
Low	507	0.40 V	0.41 V	0.01 V
Medium	1940	1.56 V	1.58 V	0.02 V
High	3999	3.22 V	3.25 V	0.03 V
Maximum (Saturation)	4017	3.23 V	3.29 V	0.06 V

- Conclusion: The system presents linear behavior with an error margin of less than 2%, which is acceptable for agricultural instrumentation and ventilation control applications.

6.3. Power Stage Validation (Boost Converter)

The DC-DC converter was subjected to a simulated load to verify output voltage stability.

- Test Condition: USB Input (5V), PWM Duty Cycle at 80%.
- Result: Stable output voltage at 12.1V.
- Thermal Observation: After 15 minutes of continuous operation, the MOSFET (Q1) and Inductor (L2) maintained temperatures below 45^{\circ}C, validating the power trace design and the selection of THT components for heat dissipation.

7. Proposed Application and Technological Extension

7.1. Evolution towards "Smart Agro-Tech"

Based on the current design, an evolution of the system towards a Smart Greenhouse Environmental Controller is proposed.

The current problem in indoor agriculture requires precise control of variables that this board is already capable of managing. The technological extension proposal focuses on three pillars:

- Full IoT Migration (MQTT/Cloud):

Leveraging the integrated ESP8266 module, the next firmware phase will transition from local UART to implementing the MQTT protocol. This will allow the board to send humidity and temperature data to a cloud dashboard (e.g., AWS IoT or Ubidots), enabling the farmer to monitor their crops from a mobile phone anywhere in the world.

- Spectral Lighting Control:

Using the Boost power stage and PWM, the proposal includes controlling LED Grow Lights. The system will automatically adjust the light spectrum:

- Vegetative Phase: Higher intensity in blue light.
- Flowering Phase: Higher intensity in red light.

This will optimize photosynthesis and reduce electrical consumption compared to fixed timers.

- Closed Loop Temperature Control (PID):

Currently, the system reads and acts directly. The proposed improvement is to implement a PID (Proportional-Integral-Derivative) control algorithm on the STM32. This will allow the greenhouse temperature to be kept constant by smoothly adjusting fan speeds instead of abruptly turning them on and off, thereby reducing thermal stress on the plants.

8. Collaboration and Team Management

To ensure the project's success within the stipulated time, a collaborative work methodology was adopted with roles defined according to the technical strengths of each member.

	Contribution
-José Eduardo Chim Cano.	Desing for IoT
-Victor Adrián Jiménez Castañeda.	Desing for Usart communication
-Alan Sebastián Ortiz Peba.	Desing for potential part
-Ian Eduardo Paredes Castillo	Desing for PCIE

Conclusion

In conclusion, the development of this custom embedded system based on the STM32F411CEU6 has successfully met all engineering objectives, effectively bridging the gap between theoretical design and physical implementation. The project resulted in a fully functional PCB that integrates power management, IoT connectivity, and control logic into a compact, industrial-grade form factor.

Experimental validation confirmed the reliability of the design, with the Boost converter delivering stable voltage and the firmware efficiently managing real-time data acquisition via ADC and telemetry via UART. Furthermore, the strict adherence to **IPC standards** (IPC-2221, IPC-7351, IPC-A-610) proved essential in ensuring manufacturability (DFM) and assembly quality, transforming a prototype design into a reliable product.

Ultimately, this project demonstrates not only the technical competency required for embedded systems design but also the ability to deliver a scalable solution ready for real-world applications in smart agriculture and industrial automation.

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